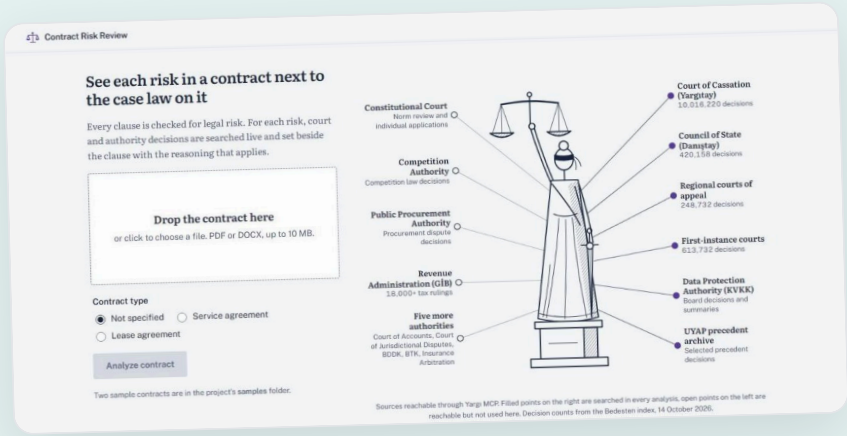
GRADUATION PROJECT · PRESENTATION 2

AI-Powered Contract Analysis System

Introduction and purpose: how the system reads a Turkish contract, scores its risks and backs them with real court decisions.



Oğuzhan Tuğay · Maksut Akgün · Eren Yüksel

Advisor: Sibel Boran · Nişantaşı University

01 THE PROBLEM WE SOLVE

People sign contracts they don’t fully understand

TODAY

- ✕ Contracts are long and full of legal wording
- ✕ A lawyer’s review is expensive for individuals and small businesses
- ✕ Unfair terms (cash payment, entry without consent) go unnoticed
- ✕ Finding the related court decisions takes legal expertise

→

WITH OUR SYSTEM

- ✓ Upload a PDF or DOCX, get a plain-language summary
- ✓ Every clause is checked and risky ones are flagged
- ✓ Each risk comes with a concrete recommendation
- ✓ Related Yargıtay, Danıştay and KVKK decisions are attached

0 – 100
overall risk score per contract

3 sources
Court of Cassation · Council of State · KVKK

10 MB
PDF / DOCX upload limit, nothing stored

02 THE RESULT

What the user gets back



1. **One risk score**
   A sample lease agreement scores **100 / 100 · Critical**, so the user sees the danger at a glance.
2. **Risks ranked by severity**
   Critical, high and medium findings are listed first, e.g. entry without consent, cash payment.
3. **Plain-language summary**
   Parties, duration, price, other provisions and dispute resolution, in a few short bullets.

ALSO INCLUDED

- Clause-by-clause findings
- Recommended actions
- Related court decisions
- Limitations of the analysis

02 THE RESULT

Every risk points to a clause

The report doesn't just say "this contract is risky". It shows the exact clause, explains why, and tells the user what to change.



ORIGINAL CLAUSE
Madde 2 · Kira bedeli, quoted in Turkish as written

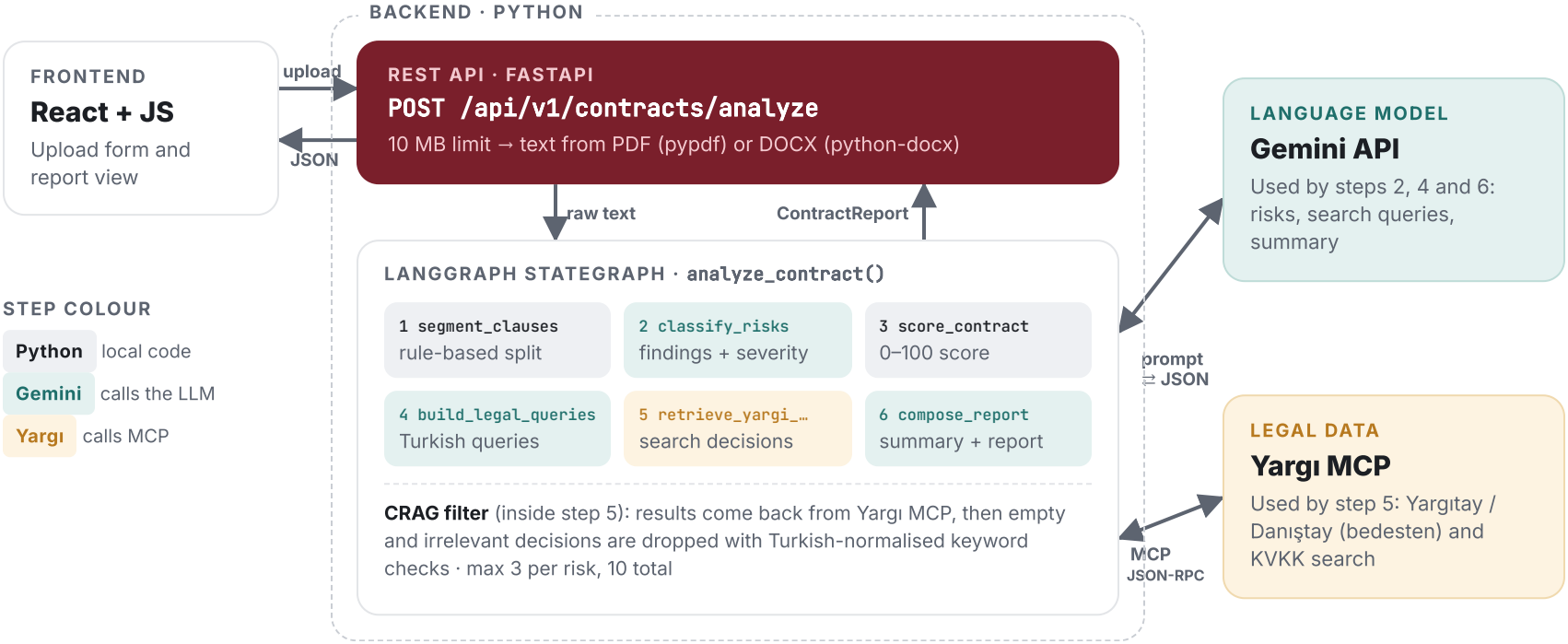
RISK + REASON
High · cash payment conflicts with banking rules

RECOMMENDATION
Require bank transfer to the lessor's account

Critical High Medium Low — Four severity levels, each with a confidence value from the model

03 SYSTEM DESIGN

Architecture: who calls whom



03 SYSTEM DESIGN

From upload to report in six steps

1

`segment_clauses`

Split into clauses

Extracted text is divided into numbered clauses by rules (MADDE headings).

Python · rule-based

2

`classify_risks`

Find risks

The model labels each risky clause with category, severity and confidence.

Gemini API

3

`score_contract`

Score 0–100

Weighted findings become one score and a risk level.

Python

4

`build_legal_queries`

Write queries

Each risk is turned into a Turkish legal search query.

Gemini API

5

`retrieve_yargi_decisions`

Fetch precedents

Decisions are searched per risk, then empty or irrelevant ones are dropped (CRAG).

Yargı MCP + CRAG

6

`compose_report`

Build the report

Gemini writes the summary; findings, actions, decisions and limitations go into one JSON.

Gemini → JSON

Each step is a node in a **LangGraph StateGraph**; they pass one shared state object (`ContractAnalysisState`) from step to step.

How the risk score is calculated

15	35	65	85
LOW	MEDIUM	HIGH	CRITICAL

```
# each finding
weight = severity_weight × confidence
# whole contract
score = max(average, highest) + density_bonus
density_bonus = min(20, 4 × (findings - 1))
```



0 · Low 25 · Medium 50 · High 90+

EXAMPLE · KIRA_SOZLESMESI.DOCX

Critical	Entry without consent	**85 × conf.**
High	Rent increase · cash · eviction	**3 × 65 × conf.**
Medium	Jurisdiction · deposit	**2 × 35 × conf.**
6 findings → density bonus		**+20**

Final score (capped at 100) **100**

Taking the highest finding means one critical clause can't be hidden by many harmless ones.

04 PURPOSE OF THE PROJECT

Why does this project exist?

To make contract review accessible: anyone should see the risks in a Turkish contract, and the court decisions behind them, before signing.

Understand

A long contract becomes a plain summary, separate clauses and one risk score from 0 to 100.

Prove

Every risk is backed by real Court of Cassation, Council of State and KVKK decisions, not only AI opinion.

Act

Each risk has a recommendation, so the user knows what to question or change before signing.

Not a replacement for a lawyer, but a first, affordable check that anyone can run.

Thank you

Questions?